

AQUAPOXY

Water-Based Epoxy Coating for Dampness Prevention

ASTRAL

TRUBUILD™

TECHNICAL DATA SHEET VER NO. N.1

PRODUCT DESCRIPTION

Trubuild Aquapoxy is a high performance, two-component epoxy resin based waterproofing compound. Part A consists of epoxy resin and Part B consists of curing agent. After mixing both the parts, it can be thinned down with water. It has good chemical resistance, adhesion, hardness, toughness and mechanical strength.

One of the major advantages of Trubuild Aquapoxy over solvent-based epoxy is that it can be applied on a damp surface also. It is non-toxic, hence eco-friendly.

FEATURES & ADVANTAGES



**EXCELLENT
RESISTANCE TO
EFFLORESCENCE**



**EXCELLENT
RESISTANCE TO
WATER INGRESS**



**COMPATIBLE
FOR MULTIPLE
SURFACES**



**CFTRI
CERTIFIED**

- Excellent vapour barrier
- Water dilutable
- Excellent porosity sealer
- Economical
- Non-toxic, hence eco-friendly
- Easy cleaning
- Anti-fungal & microbial properties
- Odourless
- Can be applied on damp surface
- Paintable
- Can exhibit positive water pressure up to 10 bar
- Can exhibit negative water pressure up to 0.5-1 bar
- When mixed with OPC white cement, cracks can be filled
- Can be used for priming concrete for PU coatings
- Can also be used on stone, tiles & asbestos sheet etc



AREAS OF APPLICATION



PACKAGING



• Underground/overhead tanks, hatcheries & aqua farms, reweries & food processing industries. • Toilet and kitchen walls, anti-efflorescence coating. • Primer before paint, roof coatings & floor coatings. • Used in food & drug industries. • Crack filling & grouting in RCC structures. • Can be used on asbestos & GI metal sheets.

METHOD OF APPLICATION

Surface preparation: Surface preparation is the first and the most important step before the application to get the results and to avoid failure. The substrate must be cleaned and free of dirt, paint, oil, grease, sludge, blistered & loose plaster, fungus & moss etc. by thorough wire brushing followed by sanding with emery paper. This is recommended all the joints. Remove the dust by compressed air, a paint brush, or wiping with a water-soaked cloth. Fill all cracks up to 3 mm by Trubuild Crackfill Paste and greater than 3mm by PMM. All the construction joints should be treated with PMM using Trubuild SBR 333.

Mixing: The Part A (Base) & Part B (Hardener) components should be mixed separately in their individual packs. The entire content of Part-B hardener should be poured into Part-A (base) & thoroughly mixed using a slow-speed stirrer/mixing paddle for 5 minutes. Then, add water in the below-mentioned ratio & mix it for another 5 minutes before application until it becomes homogeneous.

APPLICATION SURFACE	PART A	PART B	WATER BY WT.
Concrete/Cementitious	1	1	1
Tiles/Metal Sheet	1	1	0.5

Application: RCC Tank Waterproofing Application: After completing waterproofing with Aqualock Flexi on the RCC surface, followed by plastering, allow the plaster surface to cure completely. Then, apply two to three coats of Aquapoxy using a brush, with each coat applied perpendicular to the previous one at an interval of 8-10 hours. Allow the final coat to cure for at least 24 hours for hard curing.

Note: For negative-side applications, three coats are recommended. Always use a water-based exterior coating as a topcoat when applied on walls. For high-gloss tiles, use a grinder to remove the gloss before application to ensure better adhesion. Curing time may vary depending on ambient temperature. In high humidity, it may take 3-5 days for complete curing. Once fully cured, clean the surface thoroughly with plenty of water 2–3 times before storing water in the tank.

Putty Preparation: In case of persistent dampness problems use a coat of putty prepared with Trubuild Aquapoxy as directed below. Stir the contents of both base and hardener individually to achieve uniform consistency for each. Mix base and hardener together and mix thoroughly to achieve homogeneous mixture. Mix 1 part of prepared Trubuild Aquapoxy with one part of white cement and stir well to achieve a proper consistency of putty.

Efflorescence Treatment of wall: Remove all the plaster along the wall 1 foot above the top point of efflorescence on wall then clean the brick surface and its joint with fresh portable water and then apply two coats of Trubuild Aqualock flexi followed by Polymer plaster with the help of Trubuild SBR 333. Once the plaster is complete apply two to three coats of Trubuild Aquapoxy followed by any paint coating. For application of putty, it is recommended to sprinkle the sand on the last coat of Aquapoxy and let it dry for 24 hours before proceeding with putty application.

Crack filling: Open the cracks with the help of a grinder and make a V-groove. Make the surface dust-free by using water and an air blower. Then apply 1 coat of Trubuild Aquapoxy in the crack and then fill it with epoxy putty made from Trubuild Aquapoxy in the crack with a stiff consistency with approx. (B:H:Water:Cement=1;1:1:5) by wt., within half hour and leave it for 4–6 hours and then apply single coat of Trubuild Aquapoxy upon the crack and leave it for 24 hours. Note: the consistency can be changed as per requirement.

TECHNICAL DETAILS

S.NO.	PROPERTIES	STANDARD	RESULTS	TEST METHOD
1.	Appearance	Visual	Opaque	In house
2.	Colour	Visual	Off white	In house
3.	Consistency after mixing	Visual	Free flow	In house
4.	Fungal-microbial resistance	Visual	Excellent	In house
5.	Pot Life at 30°C	Minutes	Approx. 120	In house
6.	Efflorescence resistance	Visual	No yellowing / blisters	SS 500-2015
7.	Water absorption, 24 hr	%	Nil	Rilem Tube
8.	Surface Dry Time at 30°C	Minutes	Approx. 45	ASTM D 1640-2014
9.	Hard Dry time at 30°C	Hrs	24	ASTM D 1640-2014
10.	Interval between two coats (30°C)	Hrs	6-8	ASTM D 1640-2014
11.	Pull off Adhesion to Concrete (B:H:W = 1:1:1)	N/mm ²	Min. 1.5, Excellent	ASTM D 7234-2022

TECHNICAL DATA SHEET

VER NO. N.1

S.NO.	PROPERTIES	STANDARD	RESULTS	TEST METHOD
12.	Pull off Adhesion to Tiles (B:H:W = 1:1:0.5)	N/mm ²	Min. 1.5, Excellent	ASTM D 7234-2022
13.	Pull off Adhesion to Metal (B:H:W = 1:1:0.5)	N/mm ²	Min. 1.5, Excellent	ASTM D 4541-2022
14.	Depth of Water Permeability @10 Bar & Negative Hydrostatic pressure up to 3.5 Bar	mm	Nil, Nil, Pass	EN 12390-8-2019 & ASTM D 7088-2017
15.	Water Vapour Transmission	g/m ² /day	114, Excellent	ASTM E 96-2024
16.	Food Grade - CFTRI certified	mg/in ²	0.054, Pass	USFDA - 175:300, Aug-2024
17.	Chemical resistance (24 hrs)	-	-	-
18.	Alkali Solution (5% NaOH)	Visual	No colour change & no blister observed	ASTM D 1308-2020
19.	Acid Solution (5% HCl)	Visual	No colour change & no blister observed	ASTM D 1308-2020
20.	Soap Solution	Visual	No colour change & no blister observed	ASTM D 1308-2020
21.	Detergent Solution	Visual	No colour change & no blister observed	ASTM D 1308-2020
22.	Lubricant Oil	Visual	No colour change & no blister observed	ASTM D 1308-2020

COVERAGE

Approx. 70-80 sq.ft/kg/coat for waterproofing

Approx. 20-25 sq.ft/coat / 3 kg mix for wall putty

Note:

Coverage may vary due to porosity and undulation of the substrate.

In case the putty becomes stiff, a little water can be added to attain right consistency.

HANDLING PRECAUTIONS COVERAGE

Mandatory and recommended industrial hygiene procedures should be Approx. 70-80 sq.ft/kg/coat for waterproofing followed whenever our products are being handled and processed. For Approx. 20-25 sq.ft/coat / 3 kg mix for wall putty more details, please refer MSDS.

STORAGE

Container should be stored in a dry and cool area. Protect the material against direct sunlight. Storage temperature should be 10 to 30 °C. Product should remain in their original, unopened containers.

SHELF LIFE

18 months from date of manufacture when stored in a cool and dry place.

DISCLAIMER: This technical data sheet replaces all previous versions of it from the immediate effect. The information, and, in particular, the recommendations relating to the application and end-use of Astral Ltd, construction chemical products, are given in good faith based on our knowledge and experience of the products when properly stored, handled and applied under suitable favourable conditions in accordance with application instructions recommended in product technical data sheet. In practice, the differences in materials, substrates and actual site conditions are such that no warranty in respect of merchantability or of fitness for a particular purpose, nor any liability arising out of any legal relationship whatsoever, can be inferred either from this information, or from any written recommendations, or from any other advice offered. For the products on which company is offering warranty, please refer to warranty document to know entire terms and conditions. The same can be availed on request or from company website. The values mentioned in the above tables in technical data sheet are obtained in controlled lab conditions when tested properly by competent laboratory. Tolerance up to five to ten percentage on the lower side from the above values are allowable. Astral Ltd. reserves the right to change the properties of its products without notice. The end user of the product must test the product's suitability for the intended application and purpose. Astral Ltd will not bear any responsibilities on the product failure if it is wrongly positioned or if improper application methodology is adopted by the user or the applicator. The proprietary rights of third parties must be observed. All orders are accepted subject to our current terms of sale and delivery. Users must always refer to the most recent issue of the local product technical data sheet for the product concerned, copies of which will be supplied on request. No freedom from any patent owned by Seller or others is to be inferred. Because use conditions and applicable laws may differ from one location to another and may change with time, customer is responsible for determining whether products and the information in this document are appropriate for the customer's use and for ensuring that the customer's workplace, and disposal practices are in compliance with applicable laws and other governmental enactments.